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Passive Sensing Target Validation and Behavioral Forecasting in GLOBEM

Research Project Report (Studienarbeit)

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Abstract

This project tests what longitudinal passive sensing measures beyond simpler alternatives. Chronological evaluation in GLOBEM combined prior-label and temporal baselines with trajectory-clustered inference, within-person analysis, construct checks, and three model-refitting regimes.

The released binary depression-screening proxy was positive in 45.4% of training rows and 47.7% of test rows. Prior same-person positive-label rate achieved held-out AUROC 0.844; adding current-day or historical passive features produced 0.840 and 0.841, with paired increment intervals $[-0.013, 0.005]$ and $[-0.013, 0.007]$. Among 295 trajectories containing both label classes, history plus prior label achieved mean within-trajectory AUROC 0.514 (95% CI $[0.474, 0.556]$). A count-matched simulation indicated sensitivity to a homogeneous moderate effect around AUROC 0.56, while the short trajectories could not resolve subtle heterogeneous or individual-user effects. After redundant PHQ representations were excluded, cross-family psychological measures had median absolute same-day correlation 0.590 (IQR 0.251–0.722).

For tomorrow behavior, calendar-only models had mean held-out $R^2 = -0.007$ and AUROC 0.544; recent same-channel history raised these to 0.244 and 0.767, while the complete model reached 0.272 and 0.788. Full-model performance varied substantially across 16 direct targets (R^2 -0.113 to 0.637 ; AUROC 0.699 to 0.884), showing that simple behavioral persistence supplied most of the forecast value. The initial forecastability score normalized history-model error by a global target standard deviation. Its $r = -0.912$ association with person-level behavioral standard deviation exposed a specification error: it mainly ranked low-variance trajectories. A corrected comparator-relative score instead measured the benefit of history over a calendar model for the same trajectory, target, and window. Its scalar form remained comparator-dependent and was not validated as a phenotype.

A narrower post-hoc result remained: continuous profiles of comparator-relative history benefit were more similar within person-year trajectories than across matched trajectories after moment and sequence-proxy controls. The profile measures domain-specific transferability of pooled history mappings, not a learned personal trait. Cross-domain coupling and discrete routine types were unsupported. A later personalization study ended at its outcome-blind feasibility check before any model was fitted.

Data and Code Availability

Data. The report uses the de-identified GLOBEM v1.1 release described by Xu et al. [2022]. Restricted participant-level source data are not included in the submission repository. The repository contains code, analysis contracts, aggregate result files, checksums, and report figures. Source data must be obtained under the dataset’s access terms. A supplementary Android pilot used the author’s own self-instrumentation data. Exact app identities and raw personal logs are not redistributed; the report and tracked aggregate displays use generic app references.

Repository. The submission package contains the analysis code, contracts, aggregate outputs, report sources, and claim-to-output map. Participant-level GLOBEM data remain excluded under the dataset access terms.

Software. Python scripts generated the CSV, JSON, and image outputs. Report figures were regenerated from aggregate outputs only. Tectonic is the sole tested document compiler. Exact source and output paths are listed in appendix D.

AI-assisted tools. OpenAI Codex was used as an assistive tool for document structuring, code drafting, and consistency checks. The author reviewed and verified the resulting code, analyses, citations, figures, and final text and remains responsible for the submitted work.

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Abbreviations

ARI	adjusted Rand index
AUROC	area under the receiver operating characteristic curve
CI	confidence interval
IQR	interquartile range
MAE	mean absolute error
PHQ-4	four-item Patient Health Questionnaire
RAPIDS	Reproducible Analysis Pipeline for Data Streams
RQ	research question
SD	standard deviation

Chapter 1

Project Scope and Objectives

1.1 Assigned problem

Passive sensing records behavior such as movement, sleep, screen interaction, and communication. Many studies use these records to predict depression, anxiety, or other psychological labels [Onnela and Rauch, 2016, Mohr et al., 2017, Saeb et al., 2015]. Prediction performance alone does not establish that the sensor data measure the named state. Performance can also arise from stable differences between people, persistence in repeated labels, related psychological constructs, or cohort-specific patterns.

The project task was to evaluate this measurement problem in GLOBEM and to identify a defensible target for passive-sensing analysis. The project therefore separated two questions:

1. Do passive features add information about repeated psychological labels after strong non-sensor baselines are included?
2. Can directly observed behavior be predicted prospectively from temporal history, and what does a stable prediction score represent after construct controls?

The analysis included strong-baseline audits, chronological prediction tests, construct controls, robustness checks, and a feasibility-based stopping rule.

1.2 Research questions

The following are retrospective synthesis questions, defined after the experimental programme and referee-driven audits.

RQ1. Psychological-label validity. Does passive sensing improve prospective depression-label classification beyond the person’s prior label rate, and how separable are the repeated psychological outcomes?

RQ2. Behavioral forecasting. Can pooled models forecast directly observed behavior from chronological history across multiple sensing domains?

RQ3. Construct validity of forecastability. Is a scalar forecastability score stable after controlling behavioral variability and entropy-related proxies? If the scalar fails, does any continuous domain-profile result remain as a post-hoc lead?

RQ4. Mechanism and follow-up feasibility. Do cross-domain histories or discrete routine types explain the profile, and is the available data sufficient for a separately specified personalization study?

1.3 Project objectives and decision status

Table 1.1 summarizes the decision rules: *passed* and *failed* refer only to the stated project criterion, *post-hoc* identifies analyses designed after earlier results, and *stopped* denotes termination at a feasibility check before model fitting.

Table 1.1: Project objectives and final decision status. The status refers to the specific analysis in this report, not to a universal claim about passive sensing.

ID	Objective	Final status
O1	Audit psychological-label prediction against prior label history and construct overlap.	passed (boundary established)
O2	Demonstrate chronological forecasting of directly observed behaviors.	passed (mostly same-channel persistence)
O3	Validate scalar forecastability after model-artifact and construct controls.	failed (point estimate $r = 0.230 < 0.30$; CI crosses gate)
O4	Test a continuous domain profile after stronger controls.	post-hoc (internal lead only)
O5	Test cross-domain coupling and discrete behavioral types.	failed
O6	Run a frozen personalization comparison only if the data pass an outcome-blind feasibility gate.	stopped (no model run)

1.4 Deliverables and exclusions

The project deliverables are the analysis code, experiment contracts, aggregate outputs, evidence ledger, reproducibility pipeline, and this report. The report is self-contained for the main conclusions, while the repository allows detailed inspection.

The GLOBEM analyses form the primary evidence base. A separately scoped Android self-instrumentation pilot is documented in appendix C as an exploratory applied work package on baseline choice in a decision system. The separate personalization feasibility audit records the stopping decision for a proposed follow-up.

Chapter 2

Background and Related Work

2.1 Passive sensing and digital phenotyping

Digital phenotyping uses data from personal digital devices to quantify behavior and health in daily life [Onnela and Rauch, 2016]. Personal sensing research combines ubiquitous sensors, repeated measurement, and statistical learning to study mental health outside the clinic [Mohr et al., 2017]. Common modalities include GPS, physical activity, sleep, screen use, app interaction, calls, Bluetooth, and Wi-Fi.

Early work reported associations between mobility or phone-use patterns and depressive symptoms [Saeb et al., 2015]. Later work has examined the timing of these associations. Stamatis et al. [2024], for example, separated concurrent and prospective utility and distinguished within-person from between-person effects. Balliu et al. [2024] evaluated person-specific mood prediction in a longer clinical design. These studies show that target definition, time scale, population, model unit, and validation split are part of the scientific claim.

Systematic reviews report substantial heterogeneity in sensors, preprocessing, outcomes, missing-data handling, and validation designs [De Angel et al., 2022]. This heterogeneity limits direct comparison of model scores. It also makes strong baselines important. A model should be compared with the information already available from prior labels, calendar structure, and behavioral history.

2.2 Generalization and target validity

GLOBEM was created to evaluate longitudinal human-behavior modeling across years and datasets. Its benchmark reported limited generalization for multiple depression detection methods [Xu et al., 2022]. This result does not by itself identify the cause. Possible causes include population shift, sensing differences, missingness, label construction, and model overfitting.

The present project focuses on target validity. A depression classifier can perform well because a person’s previous labels are persistent. It can also benefit from stable between-person prevalence. In that case, passive features may be predictive in isolation but provide little new information. A valid incremental test must compare passive sensing with a label-history baseline using a chronological split.

Repeated psychological scales create a second issue. Depression, anxiety, stress, and affect can share variance. Some released variables may also be transformations or subscales of the same instrument. A model score against one column therefore does not establish specificity

to the named construct. The project uses empirical correlation and temporal-prior models to quantify this overlap. It does not claim that overlapping constructs are clinically meaningless.

2.3 Behavioral predictability and regularity

Behavioral predictability is an established research topic. Song et al. [2010] used entropy-related methods to estimate limits on mobility predictability. Phillips et al. [2017] proposed the Sleep Regularity Index, which measures the probability of occupying the same sleep/wake state 24 hours apart. Wang et al. [2018] studied variability, circadian structure, and flexible regularity in smartphone sensing. Ren et al. [2022] used a continuous-time hidden Markov model to estimate activity rhythms.

These studies matter for construct validation. A prediction-error score can appear stable because a person has consistently low or high behavioral variance. A profile can also reflect marginal entropy, weekday structure, or short-term autocorrelation. The present project therefore treats variability and regularity as competing explanations, not as minor covariates.

The project does not estimate a theoretical predictability ceiling. Its entropy quantities are marginal or discretized descriptive proxies. Short trajectories and discretization prevent a direct Song-style interpretation. Phrases such as “learnable beyond entropy” are therefore not supported.

2.4 Between-person and idiographic interpretation

Stable smartphone behavior can predict broad between-person characteristics. Stachl et al. [2020] reported prediction of several Big Five traits from smartphone behavior. This literature provides a useful positive-control context: a pipeline can contain stable behavioral information even if it does not measure short-term psychological states.

Group-level models and person-specific models answer different questions. Group-to-individual generalization cannot be assumed [Fisher et al., 2018]. In this project, the forecasting coefficients were pooled across eligible training rows. Relative history benefit therefore measures how well a population-average history mapping transfers to a trajectory compared with a population-average calendar mapping. It is not an idiographic model.

Dense person-specific sampling is feasible in some settings. Across seven idiographic ecological momentary assessment studies, Soyster et al. [2019] reported high prompt completion. That result supports the feasibility of future longer person-specific studies, but it does not change the model unit in GLOBEM.

2.5 Validation design

Longitudinal data violate the independence assumption behind random row splits. Validation should respect temporal, hierarchical, and cohort structure [Roberts et al., 2017]. Small or structured samples can also produce uncertain cross-validation estimates [Varoquaux, 2018]. The project therefore used chronological splits, model-refitting checks, matched within-cohort permutations, bootstrap intervals, and leave-one-cohort-out tests.

These studies motivate four requirements here: compare psychological-label models with prior-label baselines; forecast behavior prospectively; test prediction-derived constructs against variability and regularity; and distinguish pooled-model transferability from person-specific learnability.

Chapter 3

Data and Reproducibility

3.1 GLOBEM dataset

GLOBEM v1.1 contains 705 person-years from 497 people across multiple longitudinal cohorts [Xu et al., 2022]. The present analyses used four institute-year cohorts from spring academic terms. Released RAPIDS features represented screen use, physical activity, sleep, location, calls, Bluetooth, and Wi-Fi.

The released identifiers do not provide a complete cross-year person key. The unit of analysis is therefore a *person-year trajectory*. It must not be interpreted as a unique person. This distinction affects all stability and profile statements.

Different stages of the pipeline contain different row counts because their eligibility rules differ. Table 3.1 keeps these stages separate.

Table 3.1: Analysis cohorts and eligibility. RQ1 and RQ3 are separate branches; disjoint refitting and known-single-year exclusion are parallel RQ3 sensitivities rather than successive exclusions.

Stage	Recorded size	Eligibility and relation
Canonical daily table	705 trajectories	Distinct institute-year/released-ID person-year keys.
RQ1 label-baseline cohort	697 trajectories	At least four weekly depression labels; separate branch from RQ3.
RQ3 behavior-prediction input	696 trajectories	At least 42 history days and an observed future direct-behavior target.
Any RQ3 person-level score	598 trajectories	Both prospective halves for at least one scope: six of eight core or ten of sixteen rich targets.
Fixed/expanding core cohort	521 trajectories	Both halves for at least six core targets.
Disjoint-refit core sensitivity	499 trajectories	Shorter non-overlapping blocks; 22 core trajectories lost coverage.
Known-single-year sensitivity	232 trajectories	Core cohort after excluding 289 repeat-candidate trajectories.

Analysis cohorts and attrition are branched, not one linear funnel

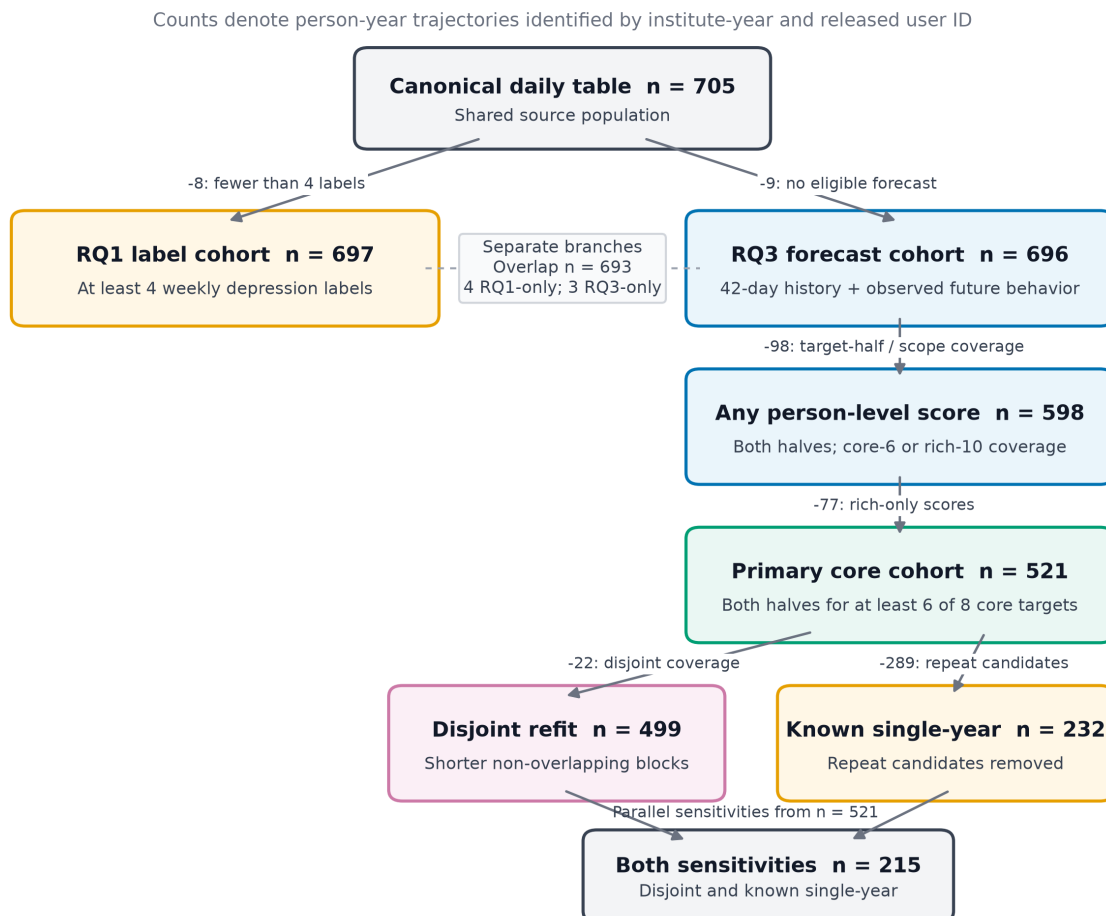


Figure 3.1: Analysis-cohort flow. RQ1 and RQ3 apply different eligibility rules and overlap on 693 trajectories; neither is nested in the other. Within RQ3, 521 trajectories form the primary core cohort. Disjoint refitting ($n = 499$) and conservative removal of repeat candidates ($n = 232$) are parallel sensitivities; their intersection contains 215 trajectories.

3.2 Behavioral and psychological variables

The direct behavior forecast attempted 16 targets derived from released aggregate features. They covered screen interaction, physical activity, sleep, mobility, communication, Bluetooth, and Wi-Fi. Continuous targets were modeled on a $\log(1+y)$ scale. Binary targets were defined relative to the training-set median.

The core construct analysis used eight relatively common targets: screen-unlock count, screen-unlock duration, steps, active duration, sedentary duration, sleep duration, time in bed, and sleep efficiency. A 16-target rich-profile sensitivity was used only as a robustness check.

The psychological audit combined ten repeated outcome columns covering depression, anxiety, stress, and affect. There were 45 possible pairwise comparisons. Twenty-nine pairs had finite correlation estimates; the remaining pairs had no or insufficient same-day overlap. Several columns are derived or closely related representations, including PHQ-4 variants. The results therefore quantify empirical overlap in the released analysis table, not ten independent diagnostic instruments.

3.2.1 Released depression-screening target

RQ1 used the released binary column `dep` from `SurveyData/dep_weekly.csv` without re-defining it. This is a screening proxy, not a clinical diagnosis, and its construction is not identical across cohorts [Xu et al., 2022, 2023]. In institute-years 2–4, weekly EMA observations were positive when the PHQ-4 total score exceeded 2, the release’s threshold for at least mild symptoms. Institute-year 1 did not administer PHQ-4; the dataset authors instead transferred a depth-2 decision rule learned in year 2 from the PANAS-style “depressed” and “nervous” items. That rule assigned a negative label only when depressed was below 2 and nervous was below 3 on their 1–5 scales. The released weekly file also merges the end-term BDI-II screening observation, whose positive threshold is greater than 13.

Consequently, `dep` is a cohort-dependent composite of weekly EMA screening and an end-term screening observation. The present analysis evaluates predictability of that released benchmark label; it does not re-estimate the instruments, harmonize their measurement properties, or infer diagnosed depression. In the frozen 65/35 within-trajectory split, 2,367 of 5,213 training rows were positive (45.4%) and 1,395 of 2,927 test rows were positive (47.7%).

3.3 Preprocessing and missing data

The analysis used released daily aggregates rather than raw sensor streams. Each experiment applied its own minimum-history and coverage requirements. Missing data were not treated as proof of behavior or state. Coverage variables were included in selected controls, and the follow-up personalization audit used count-only eligibility checks before any outcomes were analyzed.

The behavior forecast used calendar features, yesterday’s outcome, rolling 7-, 14-, and 42-day summaries, and partial-day features where applicable. Features were constructed using information available before the prediction target. The target horizons were tomorrow and the following seven-day mean.

3.4 Cross-year identity boundary

Pinned overlap metadata identified 378 repeat-candidate trajectory keys. Of the 521 core trajectories, 289 were candidates and 232 were known single-year records. The release did not expose a complete cross-year person identifier, precluding a person-cluster bootstrap. A conservative sensitivity excluded every repeat candidate; accordingly, longitudinal claims use the units “within trajectory” and “person-year profile,” not cross-year person-level stability.

3.5 Privacy and data access

The primary project is a secondary analysis of a de-identified research release. Recruitment, consent, and institutional-review procedures are described in the GLOBEM publication [Xu et al., 2022]. This report does not claim a new ethics approval. No raw GLOBEM data or participant-level follow-up tables are copied into the report folder or submission repository. The supplementary pilot is self-observation by the author and includes no third-party participant. Its raw session logs and exact app identities remain private; only aggregate counts, metrics, contracts, and de-identified displays are used in the report package.

Quantitative claims and figures trace to machine-readable CSV or JSON outputs through `CLAIM_SOURCE_MAP.md`. Figures are regenerated by `make_figures.py` from aggregate outputs, and the report package contains no restricted participant-level input data.

Chapter 4

Project Method

4.1 Staged experimental design

The project proceeded in seven stages:

1. establish passive-sensing and history baselines;
2. audit psychological labels against prior-label history and construct overlap;
3. forecast direct behavior prospectively;
4. define and test scalar and profile measures of forecast performance;
5. run construct, model-refitting, mechanism, and typology controls;
6. freeze a separate personalization design and stop it if feasibility failed; and
7. instrument and audit a separate single-user Android decision pilot.

Early results motivated later construct checks, which attributed most scalar forecastability stability to behavioral variance and changed the scientific interpretation of that score.

4.2 Chronological evaluation

For the main behavior forecast, each trajectory was split chronologically. The earlier 65% supplied the original training period, and the later 35% supplied prospective evaluation rows. No future target values were used to construct history features for an earlier prediction.

The psychological-label control also used a within-user temporal split. Its prior-label feature used expanding prior labels in training and the user's training-period label rate for test rows. A training global rate was used when a user-specific rate was unavailable.

4.3 Psychological-label baselines

Random-forest depression classifiers compared five feature sets:

1. current-day all-day passive features;
2. 7/14-day passive history;
3. prior same-person depression-label rate only;

4. current-day passive features plus prior-label rate; and
5. passive history plus prior-label rate.

The primary metric was held-out AUROC. For the two combined-versus-label-only contrasts, uncertainty was estimated by a paired 10,000-draw bootstrap over 697 released person-year trajectories; every sampled trajectory contributed all of its test rows. A 95% interval containing zero was defined as no detectable increment.

A within-person ranking test then retained only test trajectories containing both depression-label classes. AUROC was computed inside each trajectory and averaged equally across trajectories; a trajectory-constant predictor therefore scores 0.5 and cannot benefit from between-person prevalence. The primary rule required the history-plus-prior mean AUROC to reach 0.55 with a trajectory-bootstrap interval entirely above 0.5. A post-hoc design diagnostic conditioned on the observed positive and negative row counts and simulated common binormal ranking effects. It quantified the sensitivity of the aggregate mean, while the short trajectories remained insufficient for person-specific performance estimates. A separate robustness grid crossed three temporal test fractions with three random-forest initializations.

For construct overlap, all possible same-day pairs among ten repeated psychological columns were examined. The primary sensitivity excluded comparisons among four PHQ-derived representations and required at least 100 same-day observations per pair. Ridge models then predicted each target using its own prior, other constructs' priors, or both. These models quantify overlap and persistence. They are not passive-sensing models.

4.4 Prospective behavior forecasting

Six model specifications were evaluated for each target, horizon, and issue time: calendar only; recent history; a 42-day fingerprint; partial-day behavior; history plus fingerprint; and history plus fingerprint plus partial-day behavior. Continuous targets used ridge regression. Binary high/low targets used logistic regression.

The models were pooled across eligible training rows. They were not fitted separately for each person-year trajectory. Held-out R^2 and AUROC were summarized across 16 heterogeneous targets using both means and median/IQR. Neither summary represents one task, so per-target values are retained in the appendix.

4.5 Model implementation

Table 4.1 records the settings needed to interpret the headline models. Settings are reported only when they are explicit in the runner or its configuration. The project configuration does not pin the scikit-learn version; therefore a setting left to a library default is identified rather than silently treated as an immutable parameter.

Table 4.1: Headline model implementation and explicit configuration. All scaling and imputation statistics were fitted on training rows. Exact runners and checked outputs are listed in appendix D.

Stage	Estimator	Essential settings
Depression-label audit	Random-forest classifier	Median imputation; 300 trees; minimum leaf size 5; balanced class weights; all CPU cores; no feature scaling.
Construct-prior audit	Closed-form ridge regression	Training-median imputation and standardization; $\alpha = 10$; unpenalized intercept; 65/35 within-trajectory temporal split.
Behavior forecast	Ridge and logistic regression	Training-median imputation and standardization. Regression used ridge with $\alpha = 1$. Classification used balanced logistic regression with <code>liblinear</code> , 1,000 iterations, and the non-overridden scikit-learn defaults L2 penalty and $C = 1$.
Forecastability refits	Custom ridge regression	Median imputation, standardization, $\alpha = 1$, fixed/expanding/disjoint temporal regimes, 1,000 bootstrap draws, and 10,000 headline permutations.
Mechanism follow-up	Custom ridge regression	Same estimator with $\alpha = 1$, 1,000 bootstrap draws, and 10,000 headline permutations.
Typology follow-up	Spherical k -means	$k = 2, \dots, 5$; 30 initializations; 100 iterations per fold; leave-one-cohort-out assignment; 500 permutations.

4.6 Initial and corrected operationalizations of forecastability

The initial analysis operationalized forecastability for trajectory u , target k , and held-out half h as global-SD-normalized negative mean absolute error:

$$F_{ukh} = -\frac{1}{\sigma_k} \frac{1}{|I_{ukh}|} \sum_{i \in I_{ukh}} |y_i - \hat{y}_i^{\text{hist}}|, \quad (4.1)$$

where σ_k is the target’s global log-scale standard deviation. Higher values mean lower absolute error. This estimator was reproducible enough to support the initial analysis, but it was mis-specified for the intended person-relative construct. Its denominator is constant across trajectories within a target: it neither adjusts for person-level outcome scale nor asks whether history improves on a relevant comparator. Equation 4.1 is retained as the audited historical operationalization, not as a validated forecastability phenotype.

The corrected comparator-relative operationalization was

$$S_{ukh} = 1 - \frac{\text{MAE}_{ukh}^{\text{hist}}}{\text{MAE}_{ukh}^{\text{calendar}}}. \quad (4.2)$$

Both errors are evaluated for the same trajectory, target, and held-out half. S asks how much the pooled history model improves on the pooled calendar model in that context. It corrects the absence of a comparator in Equation 4.1, but remains sensitive to the calendar model, denominator error, marginal entropy, and population-average coefficients. It is not a measure of separately fitted personal-model skill.

Target-half measures were standardized within target. Equal-weight averages formed scalar degree scores. Cross-target vectors formed continuous domain profiles. Profile similarity used cosine similarity.

4.7 Decision rules and provenance

The project record specified three decision rules before the corresponding full inferential run. Because that record lacks an independent prospective timestamp, the rules organize the internal analysis but are not presented as preregistration.

- **H1 scalar stability:** early/late Pearson $r \geq 0.30$ with a trajectory-bootstrap 95% confidence interval above zero.
- **H2 profile stability:** same-trajectory early/late cosine similarity exceeds institute-year-matched cross-trajectory similarity by $\Delta \geq 0.10$ with permutation $p < 0.05$.
- **H3 trait association:** corrected association with selected traits has $|r| \geq 0.10$ and max-statistic permutation $p < 0.05$.

The numeric thresholds are project-specific rather than universal psychometric standards. The H1 rule was statistically suboptimal: it applied the practical threshold to the point estimate while testing the confidence bound only against zero. The historical decision is retained for transparency, but an interval spanning 0.30 is inconclusive about whether the population effect meets that value. A future confirmatory analysis should test uncertainty against the practical threshold.

4.8 Model-artifact and construct controls

Three temporal model regimes tested whether shared fitted coefficients produced apparent stability:

Fixed. The original pooled model scored both test halves.

Expanding refit. The model was refitted after the first test half before scoring the final half.

Disjoint refit. Separate models were trained on disjoint early and late windows and evaluated on later windows.

Construct overlap was tested against observed target standard deviation and normalized marginal quintile entropy. A stronger sensitivity estimated these moments from the corresponding training block to avoid sharing the same held-out observations with MAE. Full-trajectory sequence controls added first-order conditional entropy, weekday-conditional entropy, and a Lempel–Ziv proxy. These are descriptive controls, not theoretical entropy-rate ceilings.

4.9 Uncertainty and multiplicity

RQ1 incremental uncertainty used a paired 10,000-draw trajectory-clustered bootstrap. The within-person test also used 10,000 trajectory bootstrap draws after restricting evaluation to trajectories with both label classes. These resampling procedures preserve repeated observations inside released person-year trajectories; they do not reconstruct unavailable cross-year identities.

Split-half correlations used 1,000 trajectory bootstrap samples. Matched profile contrasts used within-cohort permutations with finite-sample correction. Headline residual profiles

used 10,000 permutations and are reported as $p < 0.001$ when they reached the resampling resolution. Secondary feature-family and original profile tests used 500 permutations; their resolution floor is $p = 0.002$.

The H3 trait test used corrected inference. A later Fisher- z approximation estimated an 80% minimum detectable absolute correlation of about 0.17 under a conservative 28-test Bonferroni approximation. This was larger than the 0.10 effect gate, so a failed H3 is treated as inconclusive rather than evidence of independence.

Feature-family decomposition and mechanism tables are explicitly post-hoc. Their individual permutation probabilities were not adjusted to support family-wise confirmatory inference; the report therefore treats them as robustness leads rather than independent discoveries.

4.10 Mechanism and typology follow-up

The post-hoc mechanism audit separated calendar terms, yesterday’s outcome, 7/14-day rolling means, a 42-day mean/standard-deviation fingerprint, all own-domain history, and recent histories from the other seven core domains. A feature family was considered an exploratory lead only if its adjusted scalar and profile gates held in all three model regimes.

Cross-domain coupling had an additional performance requirement: aggregate held-out MAE had to improve, and at least six of eight targets had to improve. Discrete typology used spherical k -means for $k = 2$ through 5. Clusters were trained on three cohorts and tested without refitting in the fourth. The typology gate required median adjusted Rand index (ARI) above 0.20, sufficient prevalence, and successful assignment stability in at least three held-out cohorts, including under disjoint refitting. With only four cohorts, this is a coarse four-trial stress test rather than strong external validation.

4.11 Personalization feasibility gate

A separate contract proposed a leave-one-institute-year-out personalization study. Phase A was outcome-blind and count-only. It required at least 30 eligible person-targets in every institute-year and at least 160 in total for selected target families. The contract required the study to stop before modeling if no allowed branch passed. This prevented a data-dependent relaxation after seeing prediction outcomes.

4.12 Implemented work packages

Table 4.2 summarizes the project sequence. On the same 196 RAPIDS history features and outer splits, the hygienically validated MLP achieved AUROC 0.619 within user and 0.559 on held-out users, compared with 0.691 and 0.577 for the random forest. A legacy LSTM on a different raw 14-day sequence reached 0.622 and 0.553; because its input and retained validation evidence differ, it is context rather than a matched comparison. Full model metrics and validation scope are reported in appendix B. Subsequent analyses therefore focused on target validity and strong non-sensor comparators rather than increasing neural capacity.

Table 4.2: Implemented work packages and decisive outcomes. Detailed runner, contract, and output paths are retained in appendix D and the repository source map.

WP	Work performed	Decisive outcome	Status
1	Baseline and model exploration	The matched-input MLP remained below the history RF; a differently specified legacy LSTM was also lower.	Mixed/null
2	Label-history and construct audit	No detectable passive increment in clustered or within-person tests; cross-family outcomes overlapped.	Primary boundary
3	Prospective behavior forecast	Direct behaviors were forecastable across 16 targets, with most performance supplied by simple same-channel history.	passed
4	Forecastability construct audit	Raw stability survived refitting but was largely variability; adjusted point estimate $r = 0.230 < 0.30$ while its CI crossed the gate.	failed
5	Post-hoc structure and typology	Continuous profile survived tested controls; cross-domain increment and discrete typology failed.	post-hoc/mixed
6	Personalization feasibility	Outcome-blind eligibility left no allowed multi-family branch; no model was fitted.	stopped
7	Android self-instrumentation pilot	Custom logger, provenance freeze, and chronological policy audit on the author's own data; observed policies were worse than no action.	post-hoc/failed

The original forecastability run passed its raw scalar and profile criteria, but the later construct audit attributed most scalar stability to behavioral variance. The separate personalization audit stopped before outcomes were examined and did not estimate a prediction effect.

Chapter 5

Results

5.1 RQ1: psychological-label validity boundary

5.1.1 Prior labels exceeded passive models

The target was the released cohort-dependent depression-screening proxy defined in Section 3.2.1. The chronological split contained 5,213 training observations (2,367 positive; 45.4%) and 2,927 test observations (1,395 positive; 47.7%). Current-day passive features achieved AUROC 0.579. Passive history achieved 0.691. The prior same-person label rate alone achieved 0.844. Current-day features plus the label rate achieved 0.840, and passive history plus the label rate achieved 0.841. Both combined models were lower than the label-only baseline (Figure 5.1 and Table 5.1). There was no positive increment beyond label history at the point-estimate level.

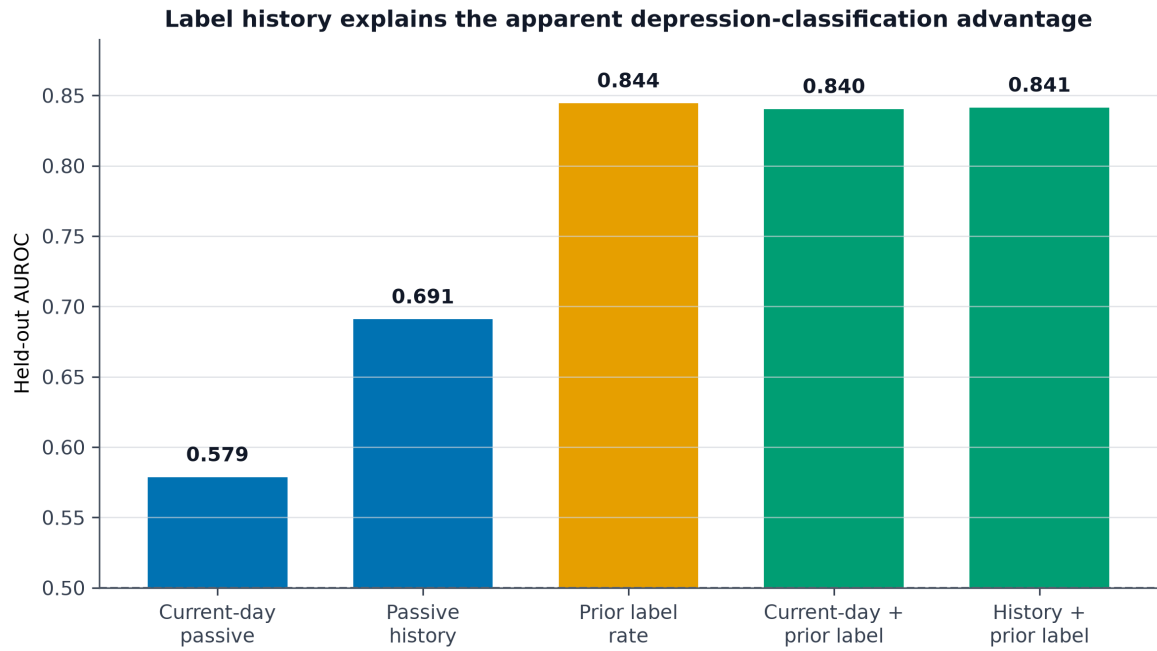


Figure 5.1: Held-out AUROC for the released depression-screening proxy in the chronological label-baseline control. The analysis used 5,213 training rows (45.4% positive) and 2,927 test rows (47.7% positive). The label-only baseline scored 0.844. Current-day plus label and history plus label scored 0.840 and 0.841, respectively. Their clustered increment intervals included zero, yielding no detectable increment over label-only. The prior-label feature is a self-report persistence/prevalence baseline, not a passive sensing model.

Table 5.1: Classification of the released depression-screening proxy. The 5,213/2,927 chronological train/test rows were 45.4%/47.7% positive. Differences were calculated from unrounded outputs and are shown to three decimals.

Feature set	AUROC	Difference vs. label-only	Interpretation
Current-day passive	0.579	−0.266	Passive-only
Passive history	0.691	−0.154	Passive-only
Prior label rate only	0.844	0.000	Strongest baseline
Current-day passive + prior label	0.840	−0.004	No positive increment
Passive history + prior label	0.841	−0.003	No positive increment

The published GLOBEM benchmark provides context but not a like-for-like reproduction. In the authors’ “Users Past/Future within One Dataset” setup, the first 80% of each user’s observations trained a model and the final 20% tested it within each dataset [Xu et al., 2022]. Its headline 0.691 ± 0.018 for Xu et al.-I is balanced accuracy, whereas 0.691 here is pooled ROC AUROC; the corresponding published mean ROC AUC was 0.737. On matched metric names, the present passive-history model was descriptively lower (balanced accuracy 0.642 versus 0.691; ROC AUROC 0.691 versus 0.737). The evaluations are not exchangeable: this study used a pooled 65/35 split, 7/14-day screen, step, and sleep history, and a fixed random forest, whereas Xu et al.-I used separate per-dataset 80/20 splits, a broader multi-epoch feature pipeline, association-rule filtering, and AdaBoost. The comparison anchors model strength without claiming a failed reproduction.

Paired trajectory-clustered inference supported the same, narrower conclusion (Table 5.2).

The current-day contrast was -0.004 with 95% interval $[-0.013, 0.005]$; the history contrast was -0.003 with interval $[-0.013, 0.007]$. Both intervals contain zero, so the result is no detectable increment, not proof that the true effects are exactly zero or negative.

Table 5.2: Paired held-out AUROC differences versus the prior-label-rate-only baseline. Intervals use 10,000 bootstrap resamples of 697 released person-year trajectories and retain all rows within each sampled trajectory.

Combined model	Δ AUROC	95% clustered CI	Decision
Current-day passive + prior label	-0.004	$[-0.013, 0.005]$	No detectable increment
Passive history + prior label	-0.003	$[-0.013, 0.007]$	No detectable increment

The within-person analysis retained 295 trajectories and 1,295 test rows with both label classes. The primary history-plus-prior model achieved an equal-weight mean within-trajectory AUROC of 0.514 (95% CI $[0.474, 0.556]$); the pair-weighted value was 0.522. Current-day-plus-prior reached 0.534 (95% CI $[0.494, 0.575]$).

Individual trajectories were sparse: the median was four test rows (IQR 4–5) and three positive-negative pairs (IQR 3–4). A post-hoc simulation preserving these class counts estimated 80% power for a common AUROC between 0.55 and 0.56, or between 0.56 and 0.57 for the complete rule that also required a point estimate of at least 0.55. Thus the aggregate analysis did not detect a moderate shared ranking effect, but it cannot exclude subtle or heterogeneous effects and cannot estimate usefulness for an individual user.

The post-hoc split/seed grid reinforced this boundary. Current-day-plus-prior and history-plus-prior were below prior-only in all nine combinations. Median differences were -0.013 (range -0.015 to -0.004) and -0.004 (range -0.012 to -0.001), respectively.

These results apply to the tested GLOBEM cohort, depression target, feature sets, and model class. Within that scope, passive sensing supplied neither a detectable aggregate increment over label history nor a detectable moderate shared within-trajectory ranking effect.

5.1.2 Psychological measures overlapped

The construct audit contained 17,346 rows with at least one psychological measure. The full released-column summary had 29 finite estimates from 45 possible same-day pairs and median absolute Pearson correlation 0.663 (IQR 0.532–0.793). That number mixes distinct constructs with redundant PHQ-derived representations. After excluding within-PHQ-family comparisons and requiring at least 100 same-day rows, 22 cross-family pairs remained; their median absolute correlation was 0.590 (IQR 0.251–0.722). Figure 5.2 shows the individual estimates and missing comparisons.

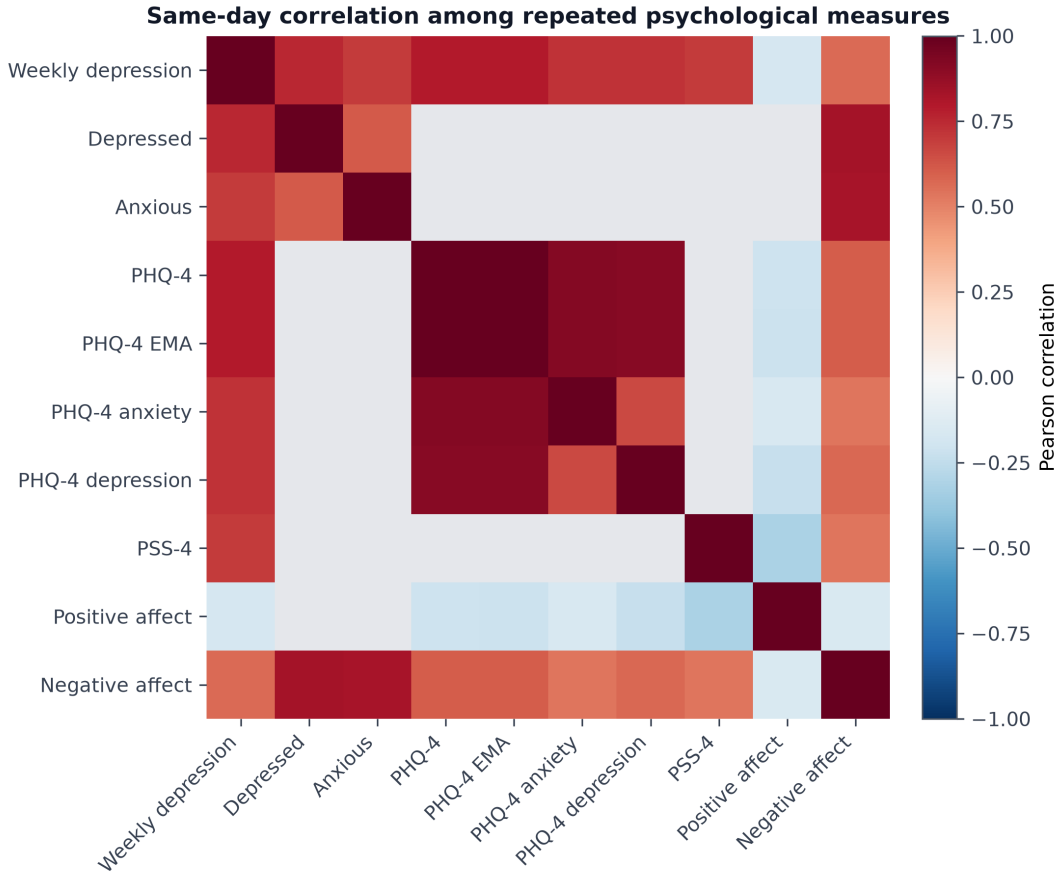


Figure 5.2: Same-day Pearson correlation among ten repeated psychological columns. Gray cells have no finite estimate because same-day overlap was absent or insufficient. Across all 29 finite pairs, the mixed median absolute correlation was 0.663. The primary nonredundant sensitivity used 22 cross-family pairs with at least 100 observations and had median 0.590 (IQR 0.251–0.722).

Mean held-out R^2 was 0.479 when each target used its own prior and 0.426 when it used only other constructs’ priors. Adding other priors to the own-prior model increased mean R^2 by 0.044. Other-construct priors approached but did not equal own-prior prediction on average. No formal equivalence test was performed.

5.2 RQ2: directly observed behavior was forecastable

Across 16 targets, the full evening model achieved median held-out tomorrow $R^2 = 0.260$ (IQR 0.225–0.323) and AUROC 0.795 (IQR 0.730–0.830). Means were 0.272 and 0.788. For the following seven-day aggregate, medians were $R^2 = 0.579$ (IQR 0.408–0.657) and AUROC 0.875 (IQR 0.829–0.920); means were 0.506 and 0.871 (Figure 5.3 and Tables 5.3 and A.1). These summaries span heterogeneous tasks and are not the performance of one target.

The baseline hierarchy materially changes the interpretation. Calendar-only models had mean tomorrow $R^2 = -0.007$ and AUROC 0.544. Adding calendar terms, yesterday’s outcome, and 7/14-day means (the recent-history comparator) raised performance to $R^2 = 0.244$ and AUROC 0.767. A calendar-plus-42-day mean/standard-deviation baseline reached 0.252 and 0.777. Combining recent and 42-day history reached 0.259 and 0.782; the current partial day supplied only the final 0.013 R^2 and 0.006 AUROC. The seven-day horizon followed the same ordering. Thus the supported result is mostly same-channel behavioral persistence, with small average increments from multiscale combination and the partial day.

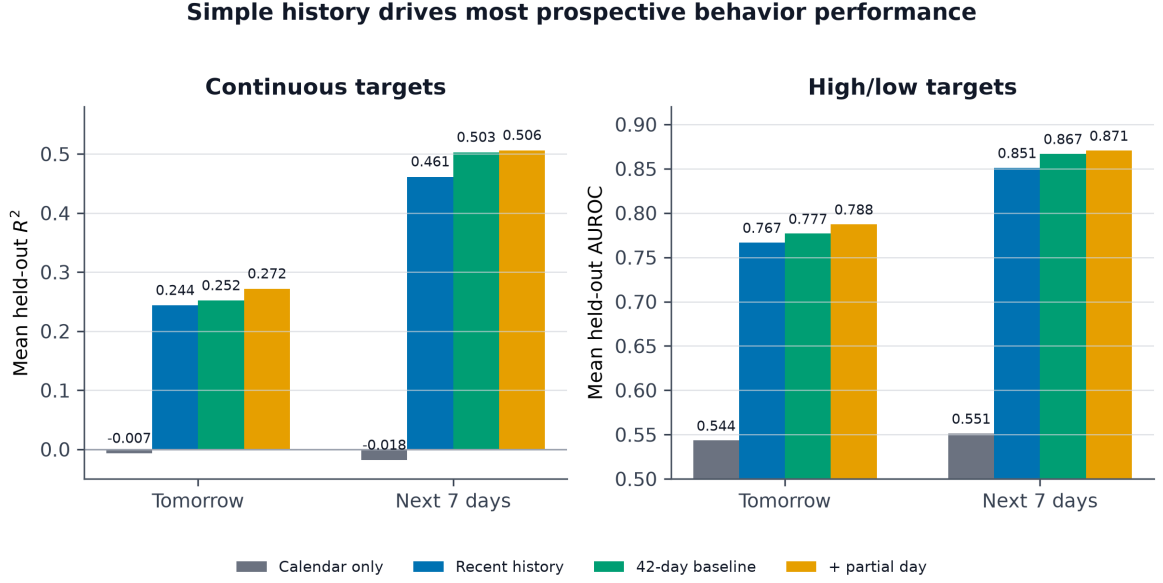


Figure 5.3: Prospective performance averaged across 16 behavioral targets at the evening issue time. Calendar-only models are followed by recent history (calendar, yesterday, and 7/14-day means), the 42-day baseline (calendar, 42-day mean and standard deviation), and the complete history/fingerprint/partial-day model. Continuous targets use held-out R^2 ; high/low targets use held-out AUROC. No common confidence interval is available for these heterogeneous target means.

Table 5.3: Mean held-out behavior-forecast performance across 16 targets at the evening issue time. Recent history contains calendar terms, yesterday, and 7/14-day means. The 42-day baseline contains calendar terms plus the 42-day mean and standard deviation. Median/IQR and full-model per-target values are reported in the text and appendix.

Horizon	Metric	Calendar	Recent history	42-day baseline	History + 42-day	+ partial day
Tomorrow	R^2	-0.007	0.244	0.252	0.259	0.272
Tomorrow	AUROC	0.544	0.767	0.777	0.782	0.788
Seven-day mean	R^2	-0.018	0.461	0.503	0.501	0.506
Seven-day mean	AUROC	0.551	0.851	0.867	0.869	0.871

Because most performance was present in the target’s own history, RQ2 establishes prospective continuity in directly observed behavior under pooled models. It does not establish that a high-dimensional cross-channel fingerprint is necessary, and it is not person-specific learning.

5.3 RQ3: correcting the forecastability operationalization

5.3.1 Raw error stability survived model refitting

The original fixed-model scalar correlation was $r = 0.412$ with a bootstrap 95% interval of $[0.316, 0.501]$ for 521 trajectories. Expanding refitting produced $r = 0.407$ with interval $[0.307, 0.496]$ for 521 trajectories. Fully disjoint refitting produced $r = 0.322$ with interval $[0.219, 0.414]$ for 499 trajectories. Shared coefficients did not fully explain the raw association.

The raw domain-profile advantage also persisted: $\Delta = 0.287$ in the original fixed analysis ($p = 0.002$, 500 permutations), $\Delta = 0.280$ under expanding refitting, and $\Delta = 0.266$ under disjoint refitting (both $p < 0.001$, 10,000 permutations). Shared fitted coefficients therefore did not fully explain the raw association.

5.3.2 Construct validation rejected the initial operationalization

At the person-average composite level, the initial score in Equation 4.1 correlated $r = -0.912$ with observed behavioral standard deviation (Spearman $r = -0.909$). Because its denominator is global rather than person-relative, the score primarily rewarded trajectories whose outcomes varied less. The proposed scalar therefore failed discriminant validity and is treated as a mis-specified operationalization rather than a newly discovered trait.

After standard-deviation and marginal-entropy adjustment, fixed-model split-half stability was $r = 0.230$ with a 95% interval of $[0.049, 0.409]$. The point estimate failed the project criterion of 0.30, while the interval remained inconclusive about whether the population correlation meets that value. Expanding and disjoint point estimates were 0.239 and 0.198. Figure 5.4 and Table 5.4 report the complete comparison.

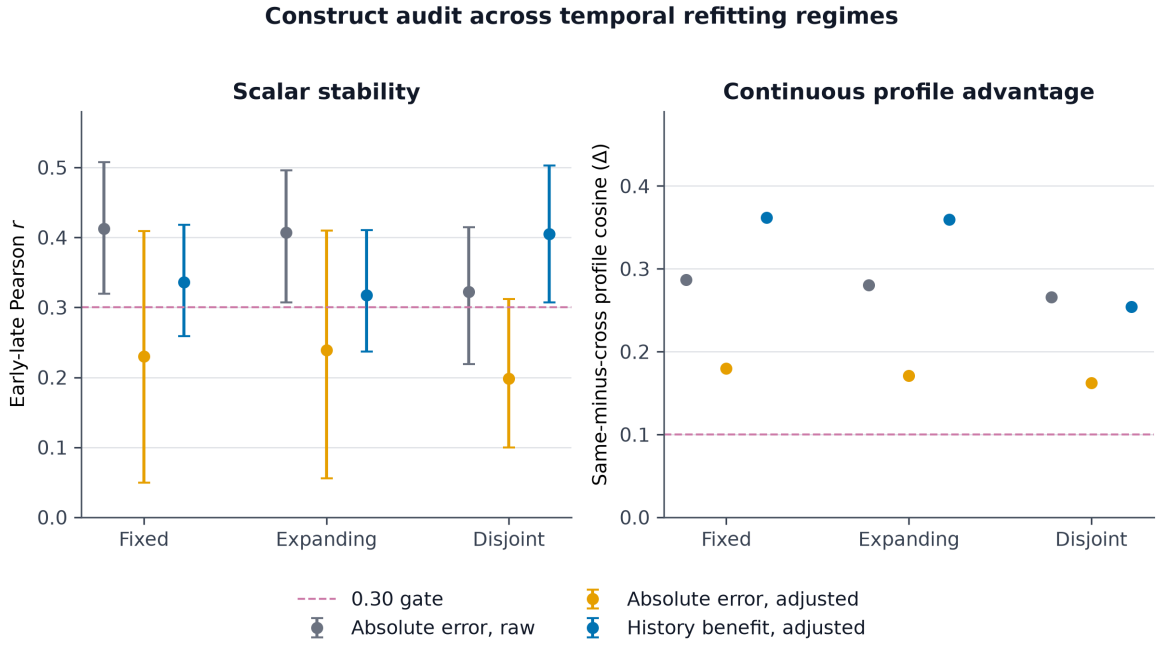


Figure 5.4: Construct audit for the core eight-domain sample. Scalar stability uses 521 trajectories in fixed and expanding regimes and 499 in the disjoint regime; error bars are trajectory-bootstrap 95% intervals. Profile advantage is the same-minus-matched-cross early/late cosine difference. The magenta lines show the project-specific 0.30 scalar criterion and 0.10 profile criterion. Adjusted rows control test-window target standard deviation and marginal quintile entropy. Profile tests used 10,000 permutations and had $p < 0.001$.

Table 5.4: Core construct-audit results. The adjusted absolute-error point estimate fails the 0.30 gate in every regime. Relative history benefit is model-relative and remains subject to the stronger controls in Table 5.5.

Measure	Regime	Degree r	95% CI	Profile Δ	Status
Absolute error, raw	Fixed	0.412	[0.316, 0.501]	0.287	Raw gate met
	Expanding	0.407	[0.307, 0.496]	0.280	Raw gate met
	Disjoint	0.322	[0.219, 0.414]	0.266	Raw gate met
Absolute error, SD/entropy adjusted	Fixed	0.230	[0.049, 0.409]	0.180	Scalar failed
	Expanding	0.239	[0.056, 0.409]	0.171	Scalar failed
	Disjoint	0.198	[0.100, 0.312]	0.162	Scalar failed
History benefit, SD/entropy adjusted	Fixed	0.336	[0.259, 0.418]	0.362	post-hoc
	Expanding	0.317	[0.237, 0.410]	0.360	post-hoc
	Disjoint	0.405	[0.307, 0.503]	0.254	post-hoc

The construct audit is a specification correction rather than a rescue of the initial scalar. Global-SD-normalized absolute error mainly measured stable behavioral variability, so it should not be named a distinct forecastability phenotype. The positive residual association is compatible with a smaller remaining signal, but it neither met the historical point-estimate rule nor isolated a distinct construct.

Equation 4.2 is the preferable operationalization for asking whether history helps relative to a calendar comparator. Under SD/entropy adjustment its scalar correlations were 0.336, 0.317, and 0.405 across fixed, expanding, and disjoint regimes. Those estimates do not validate S as a trait: stronger training-window and sequence controls attenuated the scalar, and the measure remains comparator-dependent. Only its continuous cross-domain profile is retained as a post-hoc lead.

5.3.3 The continuous profile remained a post-hoc lead

Relative history benefit was less coupled to standard deviation than raw error at the person-average level ($r = -0.268$), but it still correlated with marginal entropy ($r = -0.518$). Its raw split-half stability was 0.484, 0.473, and 0.445 under fixed, expanding, and disjoint regimes.

The scalar did not survive every stronger control. Training-window moment adjustment produced $r = 0.323$, 0.263, and 0.322; the expanding value failed the 0.30 gate. Sequence-proxy adjustment produced $r = 0.240$, 0.219, and 0.246; all failed. The history-benefit scalar is therefore not a robust phenotype.

The continuous profile was more consistent. Same-minus-cross profile advantages remained above 0.10 after three types of adjustment (Table 5.5). All listed tests used 10,000 within-cohort permutations and had $p < 0.001$.

Table 5.5: Post-hoc continuous profile of pooled relative history benefit under stronger controls. Values are same-minus-matched-cross cosine differences. Sample sizes for sequence adjustment are 439, 439, and 425 trajectories for fixed, expanding, and disjoint regimes.

Adjustment	Fixed Δ	Expanding Δ	Disjoint Δ	Status
Test-window SD and marginal entropy	0.362	0.360	0.254	post-hoc
Training-window SD and marginal entropy	0.354	0.311	0.217	post-hoc
Sequence-order proxies	0.314	0.308	0.202	post-hoc

The identity sensitivity excluded every trajectory flagged as a possible cross-year repeat. For

the core, standard-deviation/entropy-adjusted history-benefit profile, 232 fixed and expanding trajectories and 215 disjoint trajectories remained. Same-minus-matched-cross profile advantages were 0.338, 0.333, and 0.242, respectively (all $p = 0.002$, 500 permutations). This conservative exclusion shows that known repeat candidates did not drive the result, while the missing cross-year identifier still limits the unit to a person-year trajectory.

The surviving secondary result is therefore a continuous within-trajectory profile of where pooled history features transfer relative to pooled calendar features. It requires replication with recoverable identities and independent data.

5.4 RQ4: mechanism, typology, and feasibility

5.4.1 Own-domain multiscale history carried the profile

After test-window standard-deviation and marginal-entropy adjustment, 7/14-day means, their recent-history combination, the 42-day fingerprint, and all own-domain history passed the exploratory scalar and profile gates across the three regimes. Yesterday alone did not pass the scalar gate in every regime (Figure 5.5).

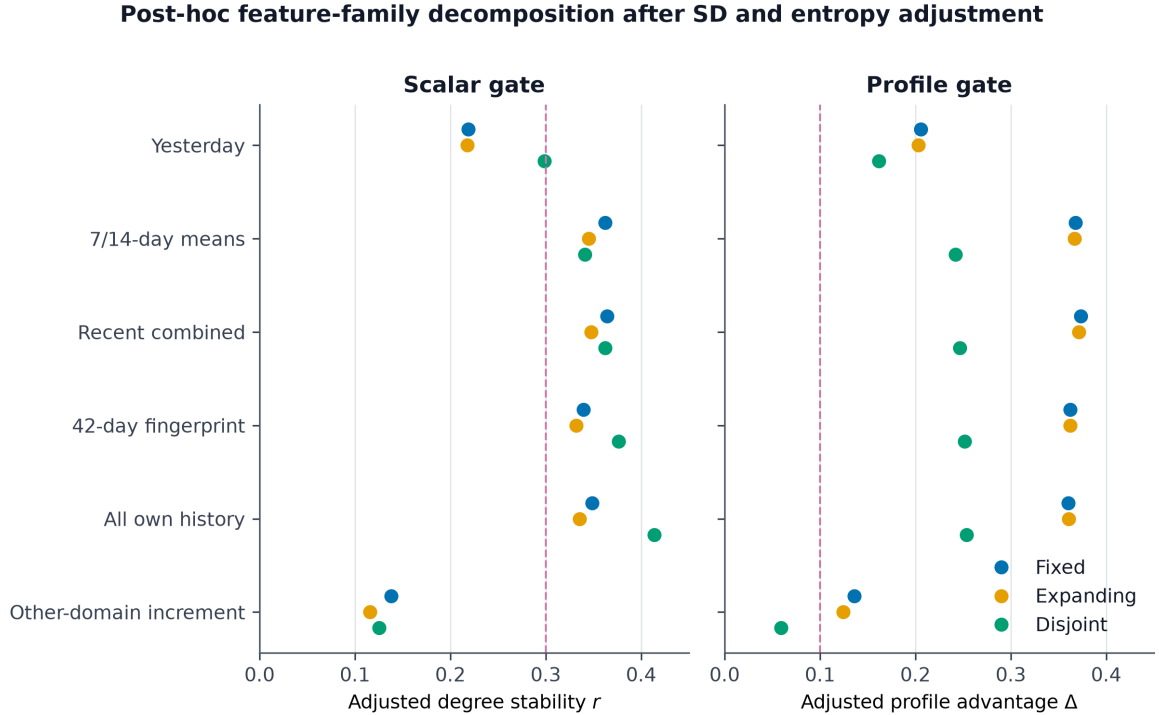


Figure 5.5: Post-hoc feature-family results after target-specific test-window standard deviation and marginal-entropy adjustment. Degree stability is early/late Pearson r ; profile advantage is the same-minus-matched-cross cosine difference. Magenta lines show exploratory 0.30 and 0.10 gates. Fixed, expanding, and disjoint core samples contain 521, 521, and 499 trajectories. Secondary profile tests used 500 permutations and reached the resolution floor $p \leq 0.002$.

The feature families overlap. Recent history and the 42-day fingerprint produced similar aggregate errors, and their combination yielded only a small additional change. The analysis does not identify one exclusive mechanism.

5.4.2 Cross-domain coupling was unsupported

Adding other-domain recent histories worsened aggregate held-out MAE by 0.47%, 0.44%, and 0.42% under fixed, expanding, and disjoint refitting. It improved four, four, and two of eight targets. The cross-domain performance gate required an aggregate improvement and at least six improved targets. It failed in every regime.

The surviving profile should therefore not be explained as cross-domain behavioral coupling. The supported description is multiscale own-domain pooled-model transferability under the tested controls.

5.4.3 No discrete routine types replicated

No k from 2 through 5 passed the leave-one-cohort-out typology gate (Figure 5.6). Median ARI ranged from 0.034 to 0.066 for fixed models and 0.028 to 0.057 for disjoint models. Every candidate had zero held-out cohorts that met the joint rule.

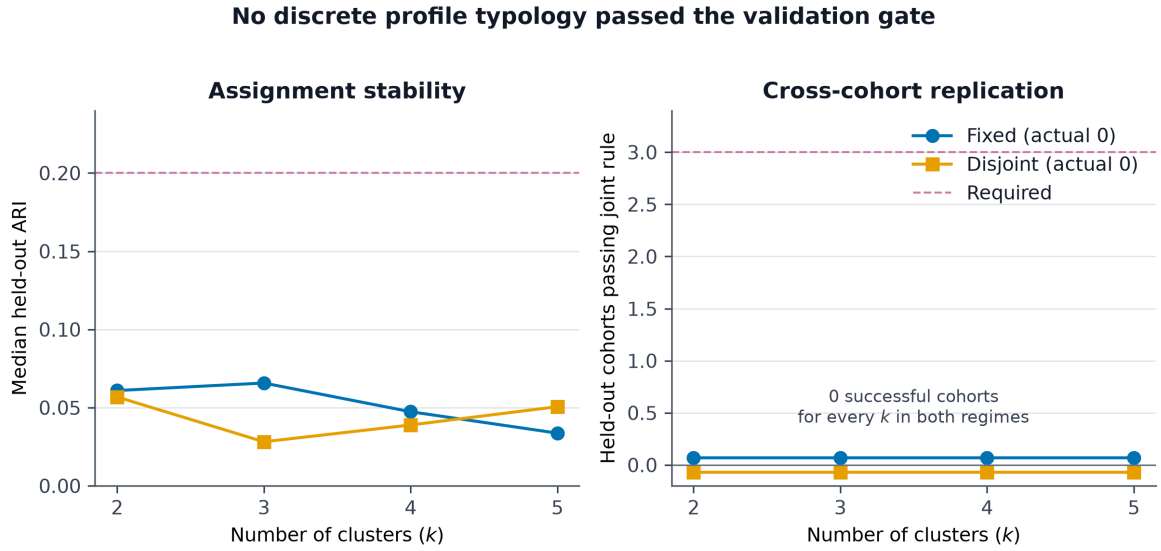


Figure 5.6: Post-hoc leave-one-cohort-out validation of spherical profile clustering. The left panel shows median adjusted Rand index across held-out cohorts; the right panel shows the number of held-out cohorts meeting the joint stability rule. Magenta lines mark the exploratory requirements of $\text{ARI} > 0.20$ and at least three successful cohorts. Both right-panel series equal zero for every k ; their markers are offset slightly only so both series remain visible. No $k = 2, \dots, 5$ passed in fixed or disjoint regimes.

Descriptive centroids can be given intuitive labels, but participants did not retain those assignments. The report therefore does not define sleep-, activity-, or screen-anchored behavioral types.

5.4.4 Personalization study stopped before modeling

The preferred branch of the outcome-blind feasibility audit retained only mobility and phone-interaction families. The selected mobility target had a minimum of 36 eligible person-targets across institute-years and 329 in total. The selected phone-interaction target also had a minimum of 36 and 327 in total. Physical activity and sleep had minima of 14 and 13, below the required 30. The reduced branch retained no family.

The contract required an allowed multi-family branch. None remained, so the selected branch and selected targets were empty. No model, prediction, loss, or outcome comparison was run.

The correct status is **stopped** at a feasibility boundary, not a personalization null.

5.5 Secondary and exploratory findings

No corrected association with conscientiousness, extraversion, or social fit passed H3. The test had limited sensitivity for effects near the 0.10 gate, so the result is **inconclusive**. It does not establish independence from personality.

The common-distress follow-up was post-hoc and cross-sectional. The original controlled estimate was $r = -0.178$ for 518 trajectories. Direct target standard deviation control gave $r = -0.140$, with cohort values ranging from approximately 0.008 to -0.342 . Because the estimates were heterogeneous and unreplicated, they are excluded from the abstract and central conclusion.

Chapter 6

Discussion and Lessons Learned

6.1 Evaluation against the research questions

RQ1 found no detectable passive increment over label history and no moderate shared within-trajectory ranking effect; sparse trajectories left subtle heterogeneous and individual effects unresolved. Cross-family outcomes still overlapped substantially. RQ2 established direct-behavior continuity under pooled chronological models; calendar-only performance was weak and simple same-channel history supplied most of the lift. RQ3 identified a specification error in global-SD-normalized forecastability and retained only a corrected, comparator-relative continuous profile as a post-hoc lead. RQ4 found no support for cross-domain improvement or discrete typology, while the separate personalization design stopped before model fitting.

6.2 Implication for passive-sensing target choice

The project supports a limited but useful conclusion. In this GLOBEM comparison, prior self-report history was more predictive of the depression label than the tested passive features. The repeated psychological columns also shared substantial variance. A passive classifier can therefore appear effective without adding new measurement of a named psychological state.

This result changes the evaluation standard. A current-day passive model should not be judged only against chance or another passive model. It should be judged against the person's available label history. A multi-construct dataset should also examine whether the target is empirically distinct from related outcomes.

Passive features may still support monitoring, context, adherence, or longer within-person studies, but a state-inference claim must first demonstrate value over the person's available label history.

6.3 What forecastability measured

The scalar audit identified a specification error. Equation 4.1 normalized absolute history-model error by a global target standard deviation. Its split-half correlation of 0.412 was reproducible under refitting, but the -0.912 association with person-level behavioral standard deviation showed that it mostly ranked low-variance trajectories. Reproducibility therefore did not confer construct validity.

Equation 4.2 corrects the principal defect by comparing history-model and calendar-model errors within the same trajectory, target, and window. This makes it appropriate for the narrower question of incremental historical information. It is still comparator-dependent,

and pooled coefficients measure transferability rather than idiographic learnability. Its scalar form also failed stronger sequence controls; only the continuous history-benefit profile remains as a post-hoc lead.

The original H1 rule adds a second lesson. The adjusted estimate of $r = 0.230$ with 95% CI [0.049, 0.409] failed the point-estimate threshold but remained compatible with effects on either side of 0.30. Future work should pre-specify an interval-based minimum effect or an equivalence/non-inferiority test.

6.4 Meaning of the continuous domain profile

The post-hoc profile result is narrower. It states that the pattern of domains in which a pooled history model outperformed a pooled calendar model was more similar within a person-year trajectory than across matched trajectories. This held after the tested moment and sequence controls.

Three boundaries are essential:

1. the models use population-average coefficients, so the profile measures transferability rather than personal-model learnability;
2. the unit is a person-year trajectory, not a cross-year identified person; and
3. the result was developed post-hoc in one dataset and requires prospective external replication.

Within these boundaries, the continuous profile is a reasonable replication target and appears to use multiscale own-domain history. No clinical or personality label is assigned to it.

6.5 Negative results as design information

The project produced several useful negative results. Richer passive context and small neural models did not rescue psychological-label prediction. Cross-domain histories did not improve the behavior forecast, and the 42-day plus partial-day terms added little on average beyond simple recent same-channel history. Discrete clusters did not replicate across cohorts. The personalization design did not have sufficient balanced data for its required multi-family comparison.

Together, these results favor simple baselines, prospective evaluation, continuous profiles, and feasibility checks before model fitting.

6.6 Applied baseline lesson

The GLOBEM analyses and the Android pilot share neither participants, features, nor models, but they expose the same evaluation error at different stages of a system. The GLOBEM psychological models required comparison with prior labels, and the behavior forecasts required temporal comparators; in the single-user pilot in appendix C, an app-based score ranked adjustment events moderately well (AUROC 0.740), yet both tested action policies produced higher error than leaving the setting unchanged. Predictive discrimination is insufficient evidence for intervention when false actions carry asymmetric cost, so policy evaluation must include a no-action baseline. A later saved-model reconstruction on the same phone showed exploratory enrichment of observed adjustment events among selected sessions, but its day-clustered interval for all-session error improvement included zero (appendix C.5).

6.7 Project-execution lessons

6.7.1 Strong baselines should be implemented early

The label-history control changed how earlier passive scores were interpreted. For longitudinal labels, prior-label and prevalence baselines belong in the first test.

6.7.2 Construct audits should follow prediction tests

Prediction and temporal reproducibility do not validate a new construct. Variance, entropy, missingness, and comparator behavior should be tested early.

6.7.3 Stopping rules protect follow-up studies

The outcome-blind personalization audit stopped an underpowered, unbalanced design before model fitting or post-hoc redefinition of its target set.

6.7.4 Confirmatory status and multiplicity

The analyses mix project-specified and post-hoc tests. Post-hoc p -values remain exploratory, not confirmatory or family-wise error-controlled.

6.8 Threats to validity

Population and season. GLOBEM contains campus cohorts observed during spring academic terms. Results may not generalize to clinical, community, older, or seasonally different populations. Shared academic calendars may contribute to behavioral continuity.

Identity. Cross-year identity is incomplete. The report cannot establish a stable person-level trait across years. Excluding repeat candidates is conservative but does not reconstruct the missing identity clusters.

Released aggregates. Models use RAPIDS aggregate features rather than raw sensor streams. Results depend on feature extraction, missingness, platform, target definition, issue time, and model class.

Time scale. Early and late windows span weeks, not years or major life changes. Stability over these windows is not long-term reliability.

Internal gates. The 0.30 and 0.10 thresholds were recorded within the project and are not universal standards. Their repository record is not independently timestamped before analysis.

Entropy proxies. Marginal quintile entropy and short discretized sequence metrics do not estimate a theoretical predictability ceiling. Their estimates are sensitive to window length and discretization.

Pooled comparator. Relative history benefit depends on both a pooled history model and a pooled calendar model. Normalized calendar MAE varied modestly across cohorts. The score is not comparator-free.

Post-hoc development and uncertainty. The profile, feature-family, sequence, typology, and distress analyses followed earlier results; their thresholds do not confer external confirmation. Aggregate behavior means have no single common confidence interval, some secondary profile tests used only 500 permutations, and H3 had limited power for small corrected associations.

Chapter 7

Conclusion and Future Work

7.1 Conclusions

This project gives four direct conclusions.

1. **Screening-label validity is the primary result.** The released target mixes PHQ-4, a year-1 PANAS-derived rule, and an end-term BDI-II observation; it is a screening proxy rather than a diagnosis. It was positive in 45.4%/47.7% of chronological train/test rows. Prior positive-label history achieved AUROC 0.844. Adding current-day or historical passive features produced 0.840 and 0.841, with clustered increment intervals that included zero. The primary within-person AUROC was 0.514 (95% CI [0.474, 0.556]). The tested passive features therefore had no detectable aggregate or moderate shared within-person increment. The available trajectories had 80% simulated power near a shared AUROC of 0.56, but remained insufficient for subtle, heterogeneous, or individual effects. Nonredundant psychological construct pairs still showed median absolute correlation 0.590.
2. **Direct behavior exhibits prospective persistence.** Calendar-only models had mean tomorrow $R^2 = -0.007$ and AUROC 0.544. Recent same-channel history raised these to 0.244 and 0.767, while the complete model reached 0.272 and 0.788. Its medians were $R^2 = 0.260$ (target range -0.113 to 0.637) and AUROC 0.795 (range 0.699–0.884). Simple history therefore supplied most forecast value; the result is continuity of observed behavior, not evidence that a rich fingerprint is necessary.
3. **The original scalar was mis-specified.** Global-SD-normalized forecastability correlated -0.912 with behavioral standard deviation and mainly ranked low-variance trajectories. Its adjusted split-half estimate was $r = 0.230$ (95% CI [0.049, 0.409]), below the project-specified 0.30 point gate. A corrected score compared history-model error with calendar-model error within the same trajectory, target, and window, but remained comparator-dependent.
4. **A continuous corrected profile remains a post-hoc lead.** The pattern of domain-specific relative history benefit survived the tested moment and sequence controls. Cross-domain coupling and discrete types were unsupported. The profile measures pooled-model transferability, not idiographic learnability or a stable cross-year phenotype.

7.2 Recommended replication study

The next study should test the continuous profile in an independent dataset with longer trajectories and recoverable person identifiers. Its contract should be timestamped before outcome analysis. The study should compare:

1. pooled history models;
2. hierarchical or mixed-effects models; and
3. separately fitted person-specific models.

The primary question should be whether the same domain profile remains within the same person across independent periods and model classes. Variability, training-window moments, missingness, weekday structure, and preselected sequence metrics should be controls from the start.

7.3 Requirements before personalization

The stopped feasibility audit should be repeated only when the candidate dataset meets its frozen per-cohort and total eligibility requirements. A revised contract may use fewer target families, but that change must be justified before inspecting model outcomes. Feasibility and effect estimation must remain separate phases.

7.4 Final project statement

The project does not support an autonomous passive detector of a named psychological state, a scalar routine phenotype, or discrete routine types. It supports a clear measurement rule: compare passive models with label history, forecast direct behavior prospectively, and audit prediction-derived scores against simpler behavioral constructs. That rule is the defensible contribution of the work.

Across the project, the resulting baseline hierarchy is: psychological-label models must beat label history and behavioral forecasts must beat temporal comparators. The separate single-user pilot in appendix C illustrates the analogous decision-system requirement that an intervention policy must beat no action.

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Appendix A

Per-Target Behavioral Forecast Results

The aggregate RQ2 summaries conceal substantial target heterogeneity. For the full evening model, tomorrow R^2 ranged from -0.113 to 0.637 and tomorrow AUROC from 0.699 to 0.884 . Table A.1 therefore reports every target for both horizons. Target names describe released aggregate channels and should not be read as psychological constructs.

Table A.1: Per-target held-out performance for the full evening model. The model combines recent history, the 42-day fingerprint, and the current partial day. Values are descriptive across heterogeneous direct-behavior targets.

Target	Tomorrow R^2	Tomorrow AUROC	Seven-day R^2	Seven-day AUROC
Sleep Efficiency	0.637	0.782	0.824	0.901
Bluetooth Unique Devices	0.561	0.882	0.633	0.935
Wifi Unique Devices	0.520	0.866	0.694	0.935
Screen Unlock Count	0.355	0.884	0.656	0.940
Outgoing Call Count	0.312	0.778	0.546	0.848
Steps Sum	0.299	0.829	0.660	0.925
Active Duration	0.265	0.811	0.645	0.903
Location Home Time	0.261	0.807	0.155	0.843
Incoming Call Count	0.259	0.747	0.530	0.846
Screen Unlock Duration	0.258	0.833	0.613	0.919
Significant Places	0.247	0.726	0.463	0.833
Location Entropy	0.243	0.732	0.385	0.809
Sedentary Duration	0.172	0.812	0.668	0.902
Sleep Duration	0.037	0.710	0.415	0.818
Sleep In Bed	0.031	0.703	0.354	0.802
Location Distance	-0.113	0.699	-0.143	0.773

Appendix B

Neural Model-Class Diagnostics

Two small neural classifiers were evaluated during model exploration. The cleanest model-class diagnostic was a tabular MLP using the same 196 RAPIDS 7/14-day history features and outer evaluation splits as the random-forest history baseline. Its inner validation respected the outer regime: later rows within each training trajectory were used for within-user validation, whereas complete user trajectories sampled within institute-year were used for held-out-user validation.

Table B.1: Single-run neural diagnostics and the random-forest history comparator. “Fit/val/test” reports row counts. Validation AUROC is absent for the random forest because it was fitted on the complete outer training block. Metrics are shown to three decimals; full-precision outputs are retained in the statistical supplement.

Model	Input	Split	Fit/val/test	Val. AUROC	Test AUROC	Macro- F1	Bal. acc.
RF	RAPIDS history	Within user	5,213/–/2,927	–	0.691	0.640	0.642
MLP	RAPIDS history	Within user	4,397/816/2,927	0.709	0.619	0.579	0.580
LSTM	Raw 14-day sequence	Within user	4,431/782/2,927	0.664	0.622	0.596	0.597
RF	RAPIDS history	Held-out users	6,063/–/2,077	–	0.577	0.548	0.548
MLP	RAPIDS history	Held-out users	5,157/906/2,077	0.547	0.559	0.536	0.536
LSTM	Raw 14-day sequence	Held-out users	5,153/910/2,077	0.698	0.553	0.541	0.542

The MLP is the matched-input diagnostic, but not an identical-training-exposure comparison: it reserved part of the outer-training pool for checkpoint selection, whereas the random forest used the complete outer-training block. It trailed the random forest by 0.071 AUROC within user and by 0.018 on held-out users. The saved LSTM run used 100 all-day sequence features plus their missingness indicators rather than the RAPIDS history vector. Its result files also predate the validation-hygiene rewrite and contain no retained split-summary artifact, so it is reported only as a legacy exploratory baseline, not as a matched or hygienic-validation comparison.

No repeated-seed distribution, neural confidence interval, or systematic hyperparameter search was run. The supported conclusion is narrow: the tested small MLP did not improve on the random-forest history baseline under the same feature representation and outer split. These runs do not establish that neural models are generally inferior for passive-sensing data.

Appendix C

Independent Android Self-Instrumentation Pilot

C.1 Scope, collection, and privacy

As a separate applied work package, the author designed, implemented, and operated a custom Android logger on their own phone. It recorded 525 listening sessions over 37 calendar days, linking generic app identity, output device, time, initial and settled media volume, and provenance evidence for matched hardware-key changes. The author was both system designer and the sole observed user, so reactivity and experimenter-participant coupling are unavoidable. This pilot is not part of the GLOBEM research questions, a transfer experiment, or a population study.

Exact app and package identities and raw session logs remain private. The report uses generic app references and only aggregate counts, metrics, contracts, and de-identified displays. No third-party participant was observed, and no suggestion or automatic actuation was enabled.

C.2 Frozen evaluation and developmental model

A protocol activation record froze the eligible collection window and evaluator. After pre-activation sessions and sessions with context changes in the evaluation window were excluded, 258 sessions remained. The first 180 sessions were used for estimation and the final 78 as a future test block; 33 test sessions contained a matched hardware-key adjustment.

The developmental two-stage mechanism was formulated after the completed prospective block and is therefore post-hoc. Stage one estimated adjustment probability from generic app identity using beta-smoothed training rates. Stage two used only adjustment-positive sessions and estimated settled volume from hierarchical training medians. Its primary specification used device and a three-hour time bucket while excluding app identity and current volume. The end-to-end policy applied the destination estimate when the stage-one probability was at least 0.50 and otherwise retained the current volume. The primary policy comparator was no action: always retain the current volume. Paired uncertainty used 2,000 session-level bootstrap resamples.

Independent Android pilot: instrumentation and locked evaluation path

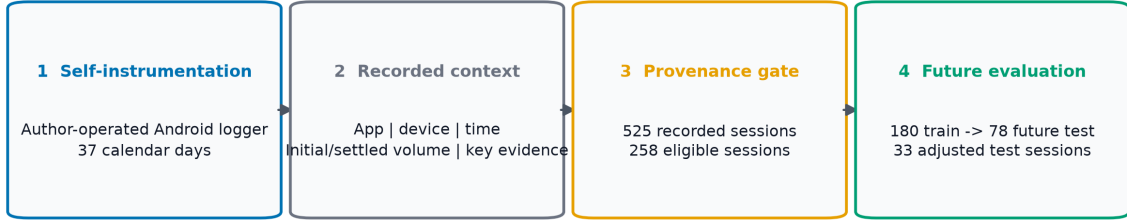


Figure C.1: Independent Android pilot workflow. The author implemented and operated the logger, recorded 525 sessions over 37 calendar days, froze a provenance-aware eligible window, and evaluated 258 eligible sessions chronologically. The pilot is a post-hoc single-user applied work package, not a GLOBEM replication or population study.

C.3 Results

Generic app identity moderately ranked whether an adjustment occurred in the 78 future test sessions: AUROC was 0.740, Brier score 0.203, and balanced accuracy 0.670 at the 0.50 threshold. The threshold produced 31 true positives and two false negatives, but also 27 false positives among 45 sessions with no observed adjustment, giving specificity 0.400. Ranking performance therefore did not by itself establish a useful action rule.

The destination-volume hypothesis failed its developmental decision rule. Device and time without app identity had MAE 21.73 on the 33 adjustment-positive test sessions, 5.70 points worse than the global median (95% paired-bootstrap interval 2.55 to 8.82). The app-aware sensitivity reached MAE 14.53, but its 1.50-point difference from the global median had interval $[-4.21, 1.50]$ and did not meet the recorded 10% practical-effect rule. Because the paired empirical bootstrap resamples only 33 adjusted sessions, its distribution is discrete; the upper 97.5th percentile equalling 1.50 is the computed quantile, not a transcription error.

Table C.1: Destination-volume MAE on the 33 adjustment-positive future test sessions. Lower is better. The app-aware sensitivity is not distinguishable from the global median under the paired bootstrap and is not reported as a positive signal.

Estimator	MAE	Interpretation
Global adjusted-session median	16.030	Constant comparator
App-aware device/time hierarchy	14.530	Difference versus median -1.50 ; 95% interval $[-4.21, 1.50]$; no supported improvement
Keep current volume	19.120	No destination estimate
Device and time, app excluded	21.730	Primary mechanism failed; paired difference from the global median was positive with interval $[2.55, 8.82]$

The complete policy was net-negative. Across all 78 sessions, no action had MAE 8.17, the observed app-aware sensitivity had MAE 13.30, and the proposed app-independent policy had MAE 17.69. Relative to no action, paired MAE differences were 5.13 points (95% interval 1.25 to 9.13) and 9.53 points (5.31 to 13.49), respectively. On the 45 unchanged sessions,

no action had MAE 0.13, whereas the app-aware and proposed policies had MAE 11.82 and 14.16. False interventions erased the descriptive ranking value.

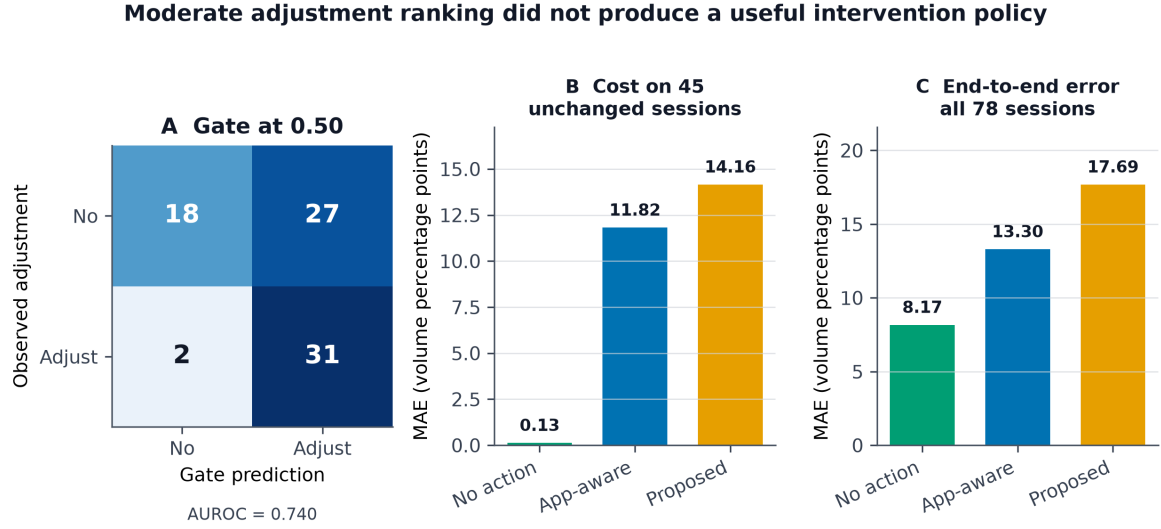


Figure C.2: Single-user Android pilot on 78 future sessions. Panel A shows the app-only adjustment gate at threshold 0.50. Panel B isolates the 45 sessions with no matched adjustment: false interventions raised MAE sharply relative to retaining the current volume. Panel C shows the same ordering end to end. “App-aware” is a descriptive sensitivity; “proposed” is the app-independent device/time destination policy. Exact app identities are omitted. Lower MAE is better.

C.4 Interpretation and limitations

The developmental two-stage mechanism and both observed policies are rejected on this block. The useful methodological result is narrower: moderate ranking performance can coexist with a harmful action policy when false positives are costly. The appropriate comparison for an intervention is therefore not only a classifier or destination baseline, but the utility of doing nothing.

This result comes from one author-user over 37 calendar days and 78 future test sessions. Hardware-key adjustment is an observed action, not a direct measure of preference, satisfaction, or intervention benefit. The mechanism was post-hoc, and there is no cross-user, platform, clinical, or deployment evidence. A future study would need multiple users, an independently frozen policy, and direct utility or preference labels before testing any actuation.

C.5 September saved-model follow-up (exploratory)

A separately dated follow-up used the same author-user’s phone export. The September 1 file held 541 sessions; the September 15 file retained those rows unchanged and added 110 sessions from September 2–15. The saved hybrid CatBoost object contains a September 1 training timestamp, a 12-point discrepancy trigger, and a 0.6 correction weight. Its feature schema uses the initial volume, app, device, time, charging/screen state, and immediately preceding-session context. Reconstructing its predictions without refitting reproduced the later reported 24 selected sessions and outcome metrics. The file was copied into the audit workspace on September 15, so the internal timestamp and matching arithmetic do not independently prove an immutable pre-September policy freeze.

Across all 110 future sessions, retaining the initial volume had MAE 9.855 and RMSE 17.899.

The saved hybrid had MAE 9.062 and RMSE 16.122, a paired MAE improvement of 0.793 points (8.0%). A day-clustered bootstrap over 14 dates gave a 95% interval $[-0.175, 1.890]$ for the improvement, which includes zero. The model changed 24 sessions. Within those sessions, MAE was 17.958 for no action and 14.324 for the hybrid; 17 changes reduced absolute error and seven raised it.

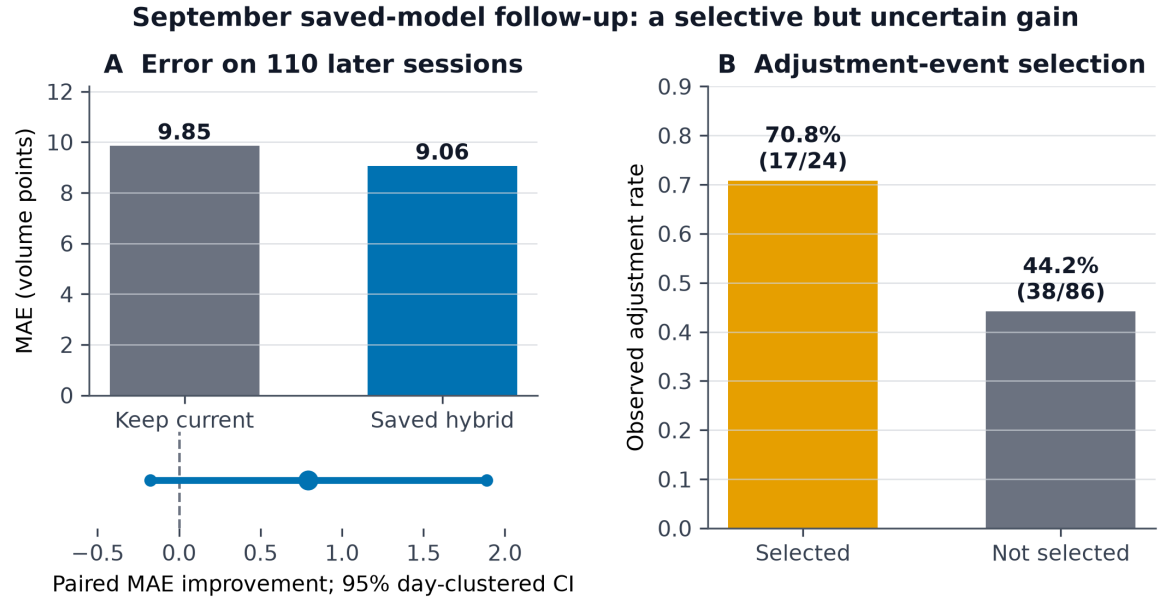


Figure C.3: Later single-user follow-up on 110 sessions from September 2–15. Panel A compares no action and the saved hybrid; the paired MAE improvement interval crosses zero. Panel B shows observed hardware-key adjustment rates for selected and unselected sessions. Selection is exploratory and post-hoc, not evidence that automatic corrections were preferred. Saved-model arithmetic was reproduced, but an immutable pre-September policy freeze was not independently verified.

Table C.2: Separate single-user blocks with different policies, not a before/after estimate for one unchanged policy. MAE covers every session; lower is better. The September interval crosses zero.

Block and policy	Keep MAE	Policy MAE	Interpretation
Developmental test: proposed two-stage policy, 78 sessions	8.170	17.690	Policy harmed overall accuracy; 95% paired difference versus keep: 5.31–13.49.
September follow-up: saved hybrid, 110 sessions	9.850	9.060	Observed 0.79-point improvement; day-clustered 95% interval: $[-0.17, 1.89]$.

The narrow positive observation is *adjustment-event selection*: 17/24 selected sessions had a hardware-key adjustment versus 38/86 unselected (70.8% versus 44.2%; difference 26.6 points, day-clustered 95% interval 3.9–50.0; uncorrected Fisher $p = 0.036$). This post-hoc, uncorrected test is exploratory, not proof that automatic correction is preferred. Six of seven selected sessions without an observed adjustment were worsened. The earlier 78-session policy null remains; neither Android result informs GLOBEM psychological labels or population efficacy.

Appendix D

Reproducibility and Source Map

Table D.1: Compact work-package source map. GLOBEM analyses have a runner and checked aggregate output; restricted source data are still required for full recomputation. Android entries are aggregate-only. Exact mappings are in the submission package’s CLAIM_SOURCE_MAP.md.

Work package	Reproducible record and represented decision
RQ1 label baseline	Runner: <code>experiments/run_label_baseline_control.py</code> ; checked output: <code>results/label_baseline_control/summary.json</code> ; target prevalence extract: <code>results/report_statistical_supplements/rq1_target_prevalence.csv</code> . Represents the label-history boundary.
RQ1 inference	Runners in <code>experiments/run_rq1_*.py</code> ; checked aggregates in <code>results/rq1_clustered_inference</code> , <code>results/rq1_within_person_deviation</code> , <code>results/rq1_within_person_power_diagnostic</code> , and <code>results/rq1_split_seed_sensitivity</code> represent clustered inference, within-person ranking, power, and split/seed sensitivity.
Construct overlap	Runners: <code>experiments/run_construct_fragility_audit.py</code> and <code>experiments/run_construct_overlap_sensitivity.py</code> ; checked outputs: <code>results/construct_fragility_audit</code> and <code>results/construct_overlap_sensitivity</code> . Represents the redundant and nonredundant construct-overlap audits.
RQ2 forecasting	Runner: <code>experiments/run_behavioural_weather_forecast.py</code> ; checked output: <code>results/behavioural_weather_forecast</code> , with the per-target reporting extract in <code>results/report_statistical_supplements/rq2_per_target.csv</code> and the baseline hierarchy in <code>results/report_statistical_supplements/rq2_baseline_summary.csv</code> .
<code>experiments/run_forecastability_phenotype.py</code> ; checked output: <code>results/forecastability/summary.json</code> . Represents the original, subsequently rejected scalar operationalization.	
RQ3 correction	Runnable: <code>experiments/run_referee_robustness_checks.py</code> ; checked output: <code>results/referee_robustness</code> . Represents the construct audit, refits, and corrected comparator-relative profile.

Continued on next page

Table D.1 continued

Work package	Reproducible record and represented decision
RQ4 structure	Runnable: <code>experiments/run_structure_mechanism_followup.py</code> and <code>experiments/run_profile_typology_followup.py</code> ; checked output: <code>results/structure_followup</code> . Represents profile mechanism and typology follow-ups.
Personalization audit	Runnable feasibility audit: <code>experiments/personalization/run.py</code> ; checked output: <code>experiments/personalization/result.json</code> . Represents the outcome-blind stop before model fitting.
Android pilot	Privacy-safe aggregate extract: <code>results/android_pilot/summary.json</code> . It verifies the displayed arithmetic and figure, not an independent rerun from private phone logs. No deployment claim.
September volume follow-up	Privacy-safe aggregate extract: <code>results/android_pilot/september_summary.json</code> . It represents a later, single-user exploratory observation. The saved-model freeze was not independently proved and the policy-improvement interval includes zero. Raw exports and predictions stay private.

D.1 Compact robustness record

Excluding all repeat-candidate trajectories retained 232 core trajectories and a raw fixed-model correlation of $r = 0.417$ with 95% interval $[0.261, 0.562]$. This shows that known repeats did not dominate the raw association; it does not validate the scalar or reconstruct unique identities. In the same exclusion sample, the standard-deviation/entropy-adjusted history-benefit profile advantages were 0.338, 0.333, and 0.242 under fixed, expanding, and disjoint regimes (all $p = 0.002$). The sequence controls were based on short discretized records (median 68 observed days) and remain descriptive. H3 was inconclusive because its approximate corrected 80% minimum detectable $|r|$ was 0.172. The post-hoc distress estimates were heterogeneous across cohorts and do not support distress prediction or within-person change detection.

D.2 Canonical build and analysis command

From `deliverables/tuhh_project_report/`:

```
python3.11 -m venv .report-venv
.report-venv/bin/pip install -r requirements-report.txt
MPLBACKEND=Agg .report-venv/bin/python make_figures.py
tectonic main.tex
```

From the repository root:

```
python3.11 -m venv .analysis-venv
.analysis-venv/bin/pip install -e .
.analysis-venv/bin/python scripts/run_rq1_clustered_inference.py
.analysis-venv/bin/python scripts/run_rq1_within_person_deviation.py
.analysis-venv/bin/python scripts/run_rq1_within_person_power_diagnostic.py
.analysis-venv/bin/python scripts/run_rq1_split_seed_sensitivity.py
.analysis-venv/bin/python scripts/run_construct_overlap_sensitivity.py
.analysis-venv/bin/python scripts/build_report_statistical_supplements.py
```

Declaration of Independent Work

I hereby declare that I prepared this report independently and used no sources or aids other than those identified in the report. All passages, data, and ideas taken directly or indirectly from other sources are marked as such and attributed to their sources. Any use of AI-assisted tools is disclosed in the Data and Code Availability statement. I reviewed the submitted analyses, code, figures, citations, and text and accept responsibility for the content of this work.

Place and date

Signature